

Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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High-temperature vacuum annealing near a metal’s melting point drives controlled grain growth but normally requires an expensive turn-key vacuum induction furnace. We describe an open, generator-agnostic modernization stack that converts a bare commercial RF induction generator into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system: LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor for metal specimens. The layer transfers between generators that expose a monotonic analog power-control input; the reproducible build is anchored on a current 6 kW solid-state generator. Validation from the archived furnace-run data and associated SEM/EBSD/optical characterization shows a fixed-geometry power–temperature calibration ($R^2 = 0.991$ over 1200–1400 °C), repeatable 1200 °C / 12 h nickel anneals (1201.2 ± 1.3 °C across 8 runs), stable soaks up to 40 h at 1325 °C, and annealed-specimen microstructures cross-referenced to their logged thermal histories. A tantalum-susceptor variant extends the furnace to non-coupling ceramics at 2000–2350 °C, demonstrated with YSZ grain growth. Complete design files, bill of materials, control software, and data archives are openly available.

Keywords: induction furnace; grain growth; vacuum annealing; LabVIEW; pyrometer; open-source hardware

I. INTRODUCTION

High-temperature annealing near a metal’s melting point drives controlled grain growth, central to studying microstructure–property relationships in metals such as nickel and iron^{1–5}. Doing this without oxidizing the sample requires both high temperature (~ 1400 – 1500 °C) and high vacuum or an inert atmosphere^{6,7}. Commercial vacuum induction furnaces meeting these requirements are closed, turn-key systems costing \$50k–\$200k+; resistive tube and box furnaces reach lower temperatures with slower ramps and offer no susceptor route to non-coupling charges; and bare industrial induction generators are widely available but ship without vacuum, optical temperature feedback, or computer setpoint control.

This work adds exactly that missing layer: a bare RF induction generator^{8,9} is retrofitted into a full annealing system with computer power control, closed-loop optical temperature feedback, a high-vacuum quartz-tube chamber, and a machined graphite crucible/susceptor. The stack is deliberately generator-agnostic: the only requirement is an analog power-control input (e.g. 4–20 mA or 0–5 V) with a monotonic command→power response, since the pyrometer feedback and PID calibration absorb any nonlinearity. The generator internals are not reproduced; a reader supplies a compatible generator and rebuilds the layer described here. The graphite crucible, which couples strongly to the RF field, serves the metal grain-growth charges; for non-coupling ceramics such as YSZ, an entirely separate tantalum-susceptor charge stack replaces it (Sec. V A).

The reproducible build documented here is anchored on a current, USA-sourced 6 kW solid-state generator

whose controller accepts a standard 4–20 mA power setpoint. The generator + heating head + controller + chiller package was quoted at $\approx \$16.5$ k, with the retrofit components itemized in the supplementary bill of materials — a small fraction of turn-key cost, in a form that is repairable and modifiable by the user. All design files, control software, run logs, and characterization data are openly available (see Data Availability), with hardware under CERN-OHL-S v2 and code under MIT (proposed).

II. SYSTEM DESIGN AND DESCRIPTION

The system comprises five subsystems (Fig. 1); Fig. 2 shows the assembled system at operating power.

The heating subsystem is a 6 kW solid-state induction generator driving a water-cooled copper work coil (~ 3 in tall, 2.5 in ID, ~ 6.5 turns, tuned to the sample/crucible stack^{10,11}); the coil conductors carry both the RF current and the coolant, and cooling water flows in series from the chiller through the generator and then the heating head/work coil before returning to the chiller. The vacuum and gas subsystem starts from the 35 mm-ID fused-quartz tube that forms the chamber, connected through a KF40 flange and flexible bellows to a compact turbo pumping station ($\sim 1 \times 10^{-6}$ – 10^{-8} Torr); a manual vent valve fed from an inert-gas regulator allows controlled venting and backfilling^{12,13}. The atmosphere matters for graphite-crucible anneals: the best results were achieved by pumping to high vacuum and then backfilling with argon through the vent valve, purging before settling to a steady trickle (~ 20 SCCM), whereas heating under vacuum alone tended to sublime graphite

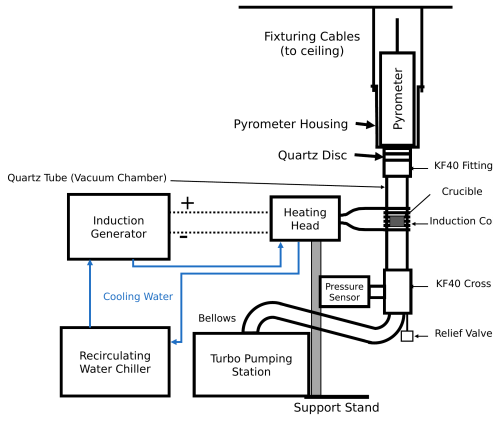


FIG. 1. System overview. The generator drives the water-cooled heating head/work coil around the vertically suspended quartz-tube chamber; the chamber stack is joined by KF40 flanges to the ceiling-suspended pyrometer housing, and an overpressure relief valve and flexible bellows lead past the pressure sensor to the turbo pumping station.

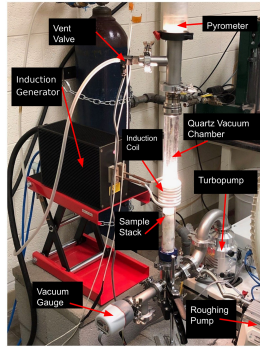


FIG. 2. The assembled, functioning furnace with the main components labeled; the quartz chamber glows at the sample-stack position, the pyrometer sights down the chamber axis, the vent valve admits the argon backfill, and the gauge, turbopump, and separately supported roughing pump complete the vacuum path.

onto the chamber walls; for the highest-temperature ceramic work (Sec. V A), the chamber was evacuated and then continuously filled with argon while the turbopump was running, avoiding oxidation during heating.

Temperature is sensed optically: a dual-wavelength (ratio) pyrometer^{14,15} (800–2500 °C sensing range) views the sample through a KF40 optical window, providing a non-contact signal robust to emissivity changes and to the RF field that would corrupt thermocouples^{16–18}. Control is implemented in LabVIEW, with VIs for manual control, automated ramping, PID tuning, and email alerts: the DAQ’s 0–5 V analog output drives a voltage-to-current loop conditioner whose 4–20 mA output lands on the generator’s power-setpoint input, and the pyrometer’s analog output returns to a DAQ input, closing the PID loop. Below the pyrometer’s ~700 °C detection floor the VI ramps open-loop;

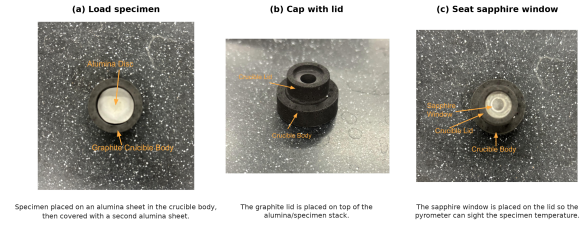


FIG. 3. Specimen-loading sequence for the graphite crucible/susceptor. (a) The specimen is placed between two alumina sheets in the crucible body. (b) The graphite lid is seated on top. (c) The sapphire window closes the lid bore, leaving an optical path for the pyrometer.

once the pyrometer returns a sustained valid reading, control hands over to closed-loop PID on the optical temperature^{19,20}. Mechanically, a bolted support stand carries the generator’s heating head, centering the work coil on the crucible position; the vacuum chamber itself has no support stand — the chamber stack is joined by KF40 flanges to the pyrometer housing, suspended from the ceiling by cables, so the vertical stack hangs from above.

Metal charges in the graphite crucible reach the Ni/Fe grain-growth range (~1400–1500 °C) with demonstrated soaks from minutes to ~40 h, and the separate tantalum-susceptor ceramic configuration extends the working range to ~2000–2350 °C. The layer transfers to any generator that exposes a monotonic analog power input, supports a water-cooled head/coil of the documented geometry, delivers enough RF power for the target charge, and allows electrical isolation of the DAQ/control path; when porting, the command scaling, calibration, PID gains, alignment, and interlock behavior are re-established — already demonstrated by moving the layer between two generators of very different designs.

A. Graphite crucible / susceptor

Metal specimens are held in a two-piece graphite crucible machined in-house from 3000 °C-grade graphite stock: a cup body and a lidded cap with a central bore that contains the charge while leaving an optical path for the pyrometer. The graphite couples strongly to the induction field and acts as a susceptor, heating the contained metal specimen by conduction and radiation. The specimen is sandwiched between two alumina discs in the cup, the lid seats on top, and a sapphire window closes the lid bore (Fig. 3); fully assembled the stack stands 13 mm tall, and the measured dimensions of every piece are given in the supplementary material. Inside the vacuum chamber the crucible rests on an alumina support rod that positions it at coil height, and crucibles are repaired with graphite cement rather than discarded. The graphite crucible is used exclusively for metal grain growth; ceramic charges are never loaded

into it, and instead use the entirely separate tantalum-susceptor stack of Sec. V A.

III. CONSTRUCTION AND OPERATION

Following the heating path, the generator’s heating head mounts on an adjustable platform on the bolted support stand so the coil centers on the crucible position of the vertically suspended quartz tube. The vacuum path begins at the quartz tube, whose open end terminates in a KF40 flange closed by a 55 mm quartz optical disc; every flanged joint compresses an elastomer O-ring carried on a centering ring, so the stack assembles with hand-tightened clamps and no tools, and these seals support the $\sim 10^{-6}$ – 10^{-8} Torr operating vacuum. From the bottom of the stack, a tee carries a 0.5 psi overpressure relief valve protecting the quartz tube during backfilling, and a flexible bellows continues through the wide-range vacuum gauge to the turbo pumping station; the roughing pump sits on a separate support, mechanically decoupled so its vibration does not reach the chamber or the pyrometer sight line (photographs in the supplementary material). On the gas path, the argon regulator feeds the manual vent valve; a mass flow controller between them is optional (a regulator and needle valve suffice). On the control path, electrical isolation is maintained between the DAQ/control wiring and the generator interface, and manual control is verified before automated ramp/soak is enabled.

Operationally, before RF power is applied the chamber must be at high vacuum or backfilled, cooling-water flow confirmed and leak-free, the coil/sample/pyrometer sight line aligned, and conductive objects kept 6–12 in from the coil. A run begins with the chamber vented (never while the turbo pump is spinning): the specimen is loaded into the crucible (Fig. 3), the stack inserted, and the chamber pumped down. The LabVIEW profile ramps open-loop below the pyrometer floor, hands over to closed-loop PID, and runs the programmed soak; afterwards the sample cools under vacuum and the chamber is vented only once the turbo pump has stopped. Off-normal conditions — loss of cooling water, vacuum loss, arcing, or runaway temperature — all call for zeroing the command and removing RF power first, then locking the system out.

IV. PERFORMANCE VALIDATION

The archive contains more than one hundred logged runs across Ni200, Ni4N5, YSZ, and Pd thermal-evaporation experiments, at ~ 900 – 1400 °C with soaks from minutes to ~ 40 h; per-run traces were parsed into uniform files and the metrics below computed reproducibly by archived scripts. Because the archive is a retrospective operating record, every comparison is restricted to an internally consistent subset; individual

runs are identified only in the supplementary specimen–run cross-reference, which links every entry to its raw log and image data.

For one fixed Ni4N5/quartz/graphite/optical configuration, four archived steady-state anneals spanning 1200 – 1400 °C give a monotonic relation between soak-mean analog power command and soak-mean pyrometer temperature, well described by $T \approx 931 + 632 I$ (°C, I in mA), $R^2 = 0.991$ (supplementary material); the broader archive shows configuration-dependent offsets, so this is an operational calibration, not a universal command–temperature law. A Ni4N5 anneal at a nominal 1300 °C / 12 h condition serves as the representative closed-loop run (trace in the supplementary material): the soak held 1302.1 ± 3.0 °C over the 12 h plateau, with a 10–90 % rise time of ~ 1135 s, overshoot of ~ 12 °C, and near-zero drift (-0.02 °C/h).

Run-to-run repeatability, assessed on the most internally consistent cohort — eight Ni4N5 anneals at a nominal 1200 °C / 12 h condition under the same gas-flow configuration — gave a soak temperature of 1201.2 ± 1.3 °C (coefficient of variation 0.11 %); runs from other periods or geometries are not pooled. Long-duration stability was demonstrated with two Ni200 anneals at 1325 °C, held for 20 h (1326.5 ± 1.5 °C, drift -0.004 °C/h) and 40 h (figures in the supplementary material). Reported temperatures are closed-loop ratio-pyrometer readings in a fixed optical geometry; absolute specimen temperature may still depend on emissivity and surface condition not fully reconstructable for every archived run^{16,21,22}.

V. MICROSTRUCTURAL VALIDATION

The furnace exists to change microstructure, so the key validation is real microstructural data tied to specific thermal histories. The archived SEM/EBSD and optical datasets are keyed by specimen IDs that map to the anneal logs; a script builds this cross-reference reproducibly (35 characterized specimens, of which 9 link to a parsed run log and 10 carry EBSD/OIM data; supplementary material). EBSD inverse-pole-figure maps of annealed specimens resolve hundreds of indexed grains, and the anneals leave surfaces that index well with no metallographic preparation: raw Kikuchi patterns show crisply resolved bands and zone axes even at coarse 8×8 binning, and more than 100,000 archived per-scan-point raw patterns permit re-indexing without re-mounting (supplementary material). Thermal grooving is itself a record of the anneal: edge-on views of a specimen annealed at 1200 °C / 12 h, cleaned only ultrasonically, show deep grooves and boundaries traversing the full sheet thickness, a step toward columnar grain structures²³.

The best-correlated specimen, Ni4N5-081 (annealed at a nominal 1300 °C for 20 h; logged soak ~ 1301 °C),

is documented across length scales in Fig. 4. Grain size is quantified with stated grain counts, metric, dispersion, and segmentation/twin-handling criteria, with the specimen as the experimental unit^{3,24}; where as-received baselines are unavailable, results are framed as final-state comparisons, and only specimens whose images link unambiguously to a logged thermal history are reported.

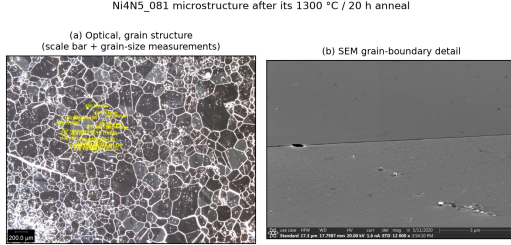


FIG. 4. Microstructure of specimen Ni4N5_081 after its logged anneal (nominal 1300°C / 20 h, gas flow). (a) Optical grain map with in-image grain-size measurements ($200\mu\text{m}$ scale bar); (b) high-magnification SEM grain-boundary detail.

A. Extension to ceramic charges: YSZ

The same control, vacuum, and sensing stack extends to charges that do not couple to the RF field at all, demonstrated with grain growth in yttria-stabilized zirconia (YSZ). Ceramic grain growth requires soak temperatures beyond the graphite/alumina stack's service range, so only the charge stack is exchanged: the YSZ specimen is sandwiched between two 25.5 mm tantalum susceptor blocks seated in a ceramic crucible (MgO or boron nitride) on an alumina support rod, while the generator, vacuum path, pyrometer, and control are untouched (schematic and micrographs in the supplementary material). With a single tantalum block the pyrometer reads $\sim 1230^{\circ}\text{C}$ at a 10 % power command, rising monotonically to $\sim 2020^{\circ}\text{C}$ at 31 %; two stacked blocks shift the curve upward, hold $2000 \pm 3^{\circ}\text{C}$ for over an hour at a constant 17 % command, and reach 2344°C at 33 % before the chamber pressure interlock ended the ramp — the practical ceiling so far is outgassing at temperature, not available RF power, and the pyrometer reads to 2500°C .

At these temperatures, chemical compatibility between specimen, susceptor, and crucible becomes the design constraint, and the archived configuration survey maps it empirically: a Ta block atop the YSZ at $2500^{\circ}\text{C}/8\text{h}$ grew grains successfully but deposited Ta vapor; YSZ set into a graphite susceptor at $2000^{\circ}\text{C}/168\text{h}$ was consumed by the carbon; and a Ta/YSZ/Ta sandwich at $2130^{\circ}\text{C}/80.4\text{h}$ formed a Ta–BN reaction shell. Figure 5 summarizes the grain growth achieved: the clean 45 min anneal at 2500°C coarsened the YSZ from ~ 20 to $\sim 90\mu\text{m}$ — growth that a commercial 1600°C box furnace needed

more than 228 h to approach (10 to $80\mu\text{m}$). Two practical rules emerged: shorter, hotter anneals stay clean (runs past $\sim 3\text{h}$ contaminated the specimen), and above $\sim 2200^{\circ}\text{C}$ a carbon susceptor not touching the YSZ is the indicated configuration.

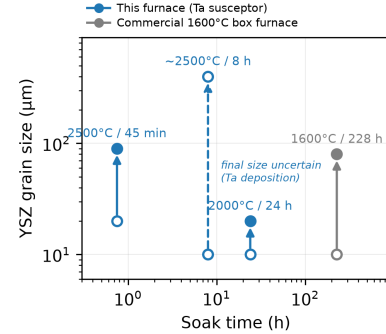


FIG. 5. YSZ grain growth in the tantalum-susceptor configuration, compared with the laboratory's reference anneal in a commercial 1600°C box furnace. Each arrow runs from the grain size before the anneal (open marker) to the grain size after it; the 8 h anneal near 2500°C is drawn dashed because tantalum deposition on the specimen made its final grain size uncertain.

VI. CONCLUSIONS

A bare commercial RF induction generator can be converted into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system with a documented, generator-agnostic modernization layer. Running on a current solid-state generator, the stack demonstrated an operational power–temperature calibration ($R^2 = 0.991$), soak repeatability of $1201.2 \pm 1.3^{\circ}\text{C}$ across eight nominally identical anneals, stable closed-loop soaks up to 40 h, and annealed-specimen microstructures cross-referenced to their logged thermal histories; a tantalum-susceptor variant extends the same furnace to non-coupling ceramics at up to 2344°C . Because the layer requires only a monotonic analog power-control input, the design is directly reusable by other laboratories at a small fraction of turn-key cost.

SUPPLEMENTARY MATERIAL

The supplementary material contains the extended figure set, the specimen–run cross-reference (Table S1, hyperlinked to the raw logs and image data), the itemized bill of materials, the design-file inventory, and the standard operating procedure; the repository and archive (see Data Availability) hold the control software, editable schematics, raw and parsed anneal logs, the characterization archives, and the scripts that reproduce every figure and metric in this paper.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

This work did not involve human or animal subjects; ethics approval is not applicable.

Author Contributions

— provisional: **Sterling G. Baird:** Conceptualization, Software, Investigation, Writing – original draft. **Ryan Weber:** Investigation, Methodology, Data curation, Visualization. **Christopher Nyborg:** Investigation, Methodology, Data curation, Visualization. **Ronnie Guymon:** Visualization, Writing – review & editing. **Gage Erickson:** Visualization, Writing – review & editing. **Oliver Johnson:** Supervision, Project administration.

DATA AVAILABILITY

The data that support the findings of this study are openly available in the development repository (<https://github.com/vertical-cloud-lab/custom-induction-furnace>); an archival snapshot is deposited at Zenodo, <https://doi.org/10.5281/zenodo.20878017>.

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